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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations
Summer Semester July 2024

CS2102-1- DATABASE MANAGEMENT SYSTEMS
(CS, IS, AI&ML, CCE, AI&DS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. What is the full form of DBMS?
A) Data of Binary Management System B) Database Management System
C) Database Management Service D) Data Backup Management System
2. What is information about data called?
A) Hyper data B) Tera data
C) Meta data D) Relations
3. _____ is a set of one or more attributes taken collectively to uniquely identify a record.
A) Primary Key B) Super key
C) Foreign key D) Candidate key
4. Which command is used to remove a relation from an SQL?
A) Drop table B) Delete
C) Purge D) Remove
5. _____ operations do not preserve non-matched tuples.
A) Left outer join B) Inner join
C) Natural join D) Right outer join
6. Which forms have a relation that contains information about a single entity?
A) 4NF B) 5NF
C) 2NF D) 3NF
7. After groups have been established, SQL applies predicates in the _____ clause, allowing aggregate functions to be used.
A) Where B) Having
C) Group by D) With
8. Which of the following resembles Create view
A) Create table ... as B) Create view as
C) Create table ... like D) With data
9. When the "ROLLUP" operator for expression or columns within a "GROUP BY" clause is used?
A) Find the groups that make up the subtotal in a row B) Create group-wise grand totals for the groups indicated in a GROUP BY clause
C) Group expressions or columns specified in a GROUP BY clause in one direction, from right to left, for computing the subtotals D) To produce a cross-tabular report for computing subtotals by grouping phrases or columns given within a GROUP BY clause in all available directions

10. Consider the relation schema: Singer(singerName, songName). What is the highest normal form satisfied by the "Singer" relation schema?
 A) 1NF B) 2NF
 C) BCNF D) 3NF
11. Which of the following is used to denote the selection operation in relational algebra?
 A) Pi (Greek) B) Lambda (Greek)
 C) Omega (Greek) D) Sigma (Greek)
12. Which is a join condition contains an equality operator:
 A) Equijoins B) Cartesian
 C) Natural D) Left
13. Which of the following makes the transaction permanent in the database?
 A) View B) Commit
 C) Rollback D) Flashback
14. A relation in which every non-key attribute is fully functionally dependent on the primary key and which has no transitive dependencies is said to be in
 A) BCNF B) 2NF
 C) 3NF D) 4NF
15. In order to maintain the consistency during transactions, database provides
 A) Commit B) Atomic
 C) Flashback D) Retain
16. _____ will undo all statements up to commit?
 A) Transaction B) Flashback
 C) Rollback D) Abort
17. Materialized views make sure that
 A) View definition is kept stable B) View definition is kept up-to-date
 C) View definition is verified for error D) View is deleted after specified time
18. The _____ is essentially used to search for patterns in target string.
 A) Like Predicate B) Null Predicate
 C) In Predicate D) Out Predicate
19. Which of the following is the syntax for views where v is view name?
 A) Create view v as "query name"; B) Create "query expression" as view;
 C) Create view v as "query expression"; D) Create view "query expression";
20. The number of attributes in relation is called as its
 A) Cardinality B) Degree
 C) Tuples D) Entity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Briefly explain the various characteristics of the database approach.	8	L*1	1	1
	b) Explain the following Relational Algebra operations with example for each:				
	i. Division				
	ii. Union				
	iii. Cartesian product				
	iv. COUNT aggregate function	8	L2	2	1,2
2.	a) Design an ER diagram and schema for the flight management system. Specify any assumptions made.	8	L3	1	1
	b) Explain grouping concepts in relational algebra. Give one example.	4	L2	2	1,2
	c) Explain following aggregate functions used in relational algebra. Give example for each.				
	i. AVERAGE				
	ii. COUNT	4	L2	2	1,2

3. a) With a neat block diagram explain the three schema architecture for databases. 8 L1 1 1
- b) Consider the Sailors-Boats-Reserves DB described Below
 SAILOR (sid, sname, rating, age)
 BOAT (bid, bname, color)
 RESERVES(sid, bid, date)
 Write each of the following queries in Relational Algebra.
- Find all sailor id's of sailors who have a rating of at least 8 or reserved boat 103.
 - Find the names of sailors who have reserved a red and a green boat.
 - Find the sids of sailors with age over 20 who have not reserved a red boat.
- 8 L3 2 1

Unit – II

4. a) Consider the following relation schema:
 Employee (employee-name, street, city)
 Works (employee-name, company-name, salary)
 Company (company-name, city)
 Manages (employee-name, manager-name)
- Answer the following queries:
- Find the names of all employees in the database who do not work for 'First Bank Corporation'. Assume that all people work for exactly one company.
 - Find the names of all employees in the database who earn more than every employee of 'Small Bank Corporation'. Assume that all people work for at most one company.
- 8 L3 3 1,2
- b) What is a view? Give the specification of view in SQL. 8 L1 4 1,2
5. a) Consider the Relation Schema $R = (A, B, C, D, E, G)$ and the FD set $F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$. Discover whether the following binary decomposition of R has the lossless join property with respect to F.
 $R_1 = \{A, B, C\}$ and $R_2 = \{A, C, D, E, G\}$
 $R_1 = \{A, B, C, D, E\}$ and $R_2 = \{A, D, G\}$ 8 L3 4 1,2
- b) How basic constraints are specified in SQL-99. Explain with example. Give syntax. 8 L1 3 1,2
6. a) Consider the following relations for an order processing database application in a company:
- CUSTOMER (cust #: int, cname: string, city: string)
 ORDER (order #: int, odate: date, cust #: int, ord-Amt: int)
 ORDER – ITEM (order #: int, item #: int, qty: int)
 ITEM (item #: int, unit price: int)
 SHIPMENT (order #: int, warehouse#: int, ship-date: date)
 WAREHOUSE (warehouse #: int, city: string)
- Develop SQL queries for the following:
- Produce a listing: CUSTNAME, #oforders, AVG_ORDER_AMT, where the middle column is the total numbers of orders by the customer and the last column is the average order amount for that customer.
 - For each item that has more than two orders, list the item, number of orders that are shipped from at least two warehouses and total quantity of items shipped.
- 8 L3 3 1,2

- b) What is the need for normalization? Explain first, second and third normal form by giving example for each.

8 L2 4 1,2

Unit – III

7. a) Explain Strict Two-Phase Locking protocol in detail.
 b) Explain the following index data structures in detail:
 i. Hash based indexing
 ii. Tree based indexing

8 L2 5 1

8 L2 5 1

8. a) Illustrate the anomalies associated with Interleaved Execution. Explain in detail.
 b) Give the algorithm to insert an element in a B+ tree.

8 L2 5 1

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
